AE 623 Project Report Template

First Author, Second Author, Third Author, Fourth Author, and Fifth Author
Group Name

December 22, 2025

This template presents the format of a technical report for AE 623. The abstract should summarize the problem, approach, and key results. It should be one paragraph.

1 Introduction

Technical reports must not exceed 10 pages in length, using the font size and margins specified in this template. Figures, references, appendices, code snippets, and effort breakdown all count towards the 10 page limit. You are encouraged to use Latex and this template. You are allowed to use other typesetting tools, e.g. Word, but you must emulate this template as closely as possible.

The introduction should state the problem, any relevant literature background, and a brief overview of the solution approach. Citations should be hyperlinked and referred to by number only [1], unless starting a sentence. Ref. [1] presents one of the first discontinuous Galerkin implementations.

2 Methods

Describe the methods you use to solve the given tasks, and include key derivations, or outlines thereof, as needed. Be complete, but avoid repetition and lengthy derivations – you do not need to demonstrate your ability to perform routine mathematical tasks, such as applying the chain rule, or solving/manipulating algebraic equations. Do not include a nomenclature, and instead define the symbols as you use them.

2.1 Governing Equations

We consider the Euler equations of gas dynamics,

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot \vec{\mathbf{F}} = \mathbf{0}, \tag{1}$$

where in two dimensions, the conservative state is $\mathbf{u} = [\rho, \rho u, \rho v, \rho E]^T$, and $\vec{\mathbf{F}} = \mathbf{F}\hat{x} + \mathbf{G}\hat{y}$ is the flux vector. While Eqn. 1 is unsteady, in this project we are only concerned with the steady-state solution.

2.2 Algorithms and Code

Discuss your high-level code structure, including memory layout, key loops, initialization, user interface, run scripts, post-processing, and any trade-offs between storage and computation. You can present algorithms, or portions thereof, in pseudocode. Small code snippets, such as the one given in Listing 1, may be useful, but do not include large code sections. Make sure your code, which you need to include in the `.zip` archive, is neat and commented. Describe any tests you performed on your code.

Listing 1: Sample code snippet

```
# print the text in quotes
print('Hello World')

# print the text in quotes 10 times
for i in range(10):
    print('Hi')

# print the numbers 0 to 9
for i in range(10):
    print(i)
```

3 Results

Present the results requested in the project statement, broken down by task or by other criteria, as you deem appropriate. Ensure that figures are of high resolution and that plots are labelled and captioned. Include a discussion of the results.

3.1 Task 1

Figure 1 shows the computational mesh and Mach number contours for a 5th-order simulation of inviscid flow over a multi-element airfoil.

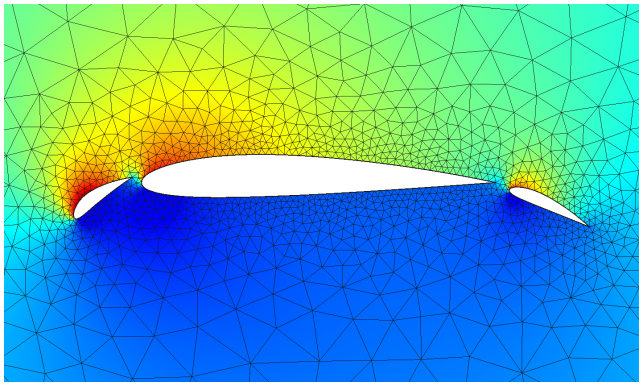

Figure 1: Mach contours (0-0.7) for a multi-element airfoil at free-stream Mach number 0.2 and angle of attack $\alpha = 10°$.

3.2 Task 2

Table 1 presents a sample table.

Table 1: Lift coefficient data.

Mesh	# cells	order	c_ℓ
coarse	2000	1	0.53
medium	8000	1	0.57
medium	8000	2	0.61

4 Conclusions

Summarize the problem, tasks, results, and draw take-away conclusions.

5 Effort Breakdown

Write a few sentences explaining how each group member contributed to the project, and include Table 2 – use initials for the column headers, and make sure each row adds up to 100%. Development refers to research, analysis, derivations, and code planning. Coding is writing of the computer code. Examples of managing are coordinating meeting times, assigning tasks, setting up shared repositories or documents, etc. Report refers to writing and formatting the report. The effort breakdown does not need to be uniform, and grades will only be affected in cases of severe and consistent disparities.

Table 2: Effort percentages.

	1A	2A	3A	4A	5A
development	50	10	10	15	15
coding	20	35	15	15	15
managing	10	10	10	60	10
report	20	20	20	20	20

References

[1] W. Reed and T. Hill. Triangular mesh methods for the neutron transport equation. Los Alamos Scientific Laboratory Technical Report LA-UR-73-479, 1973.